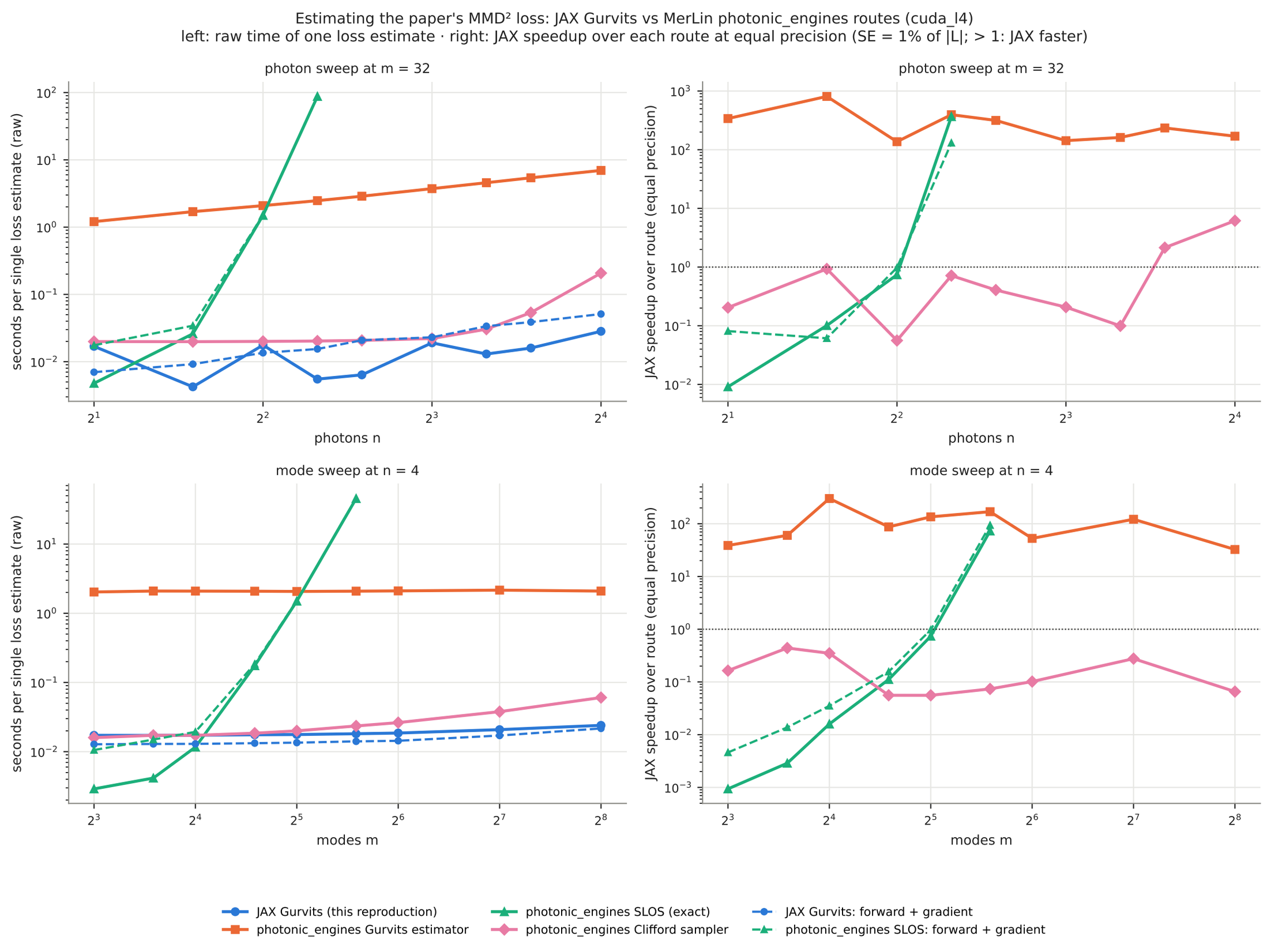
Estimating the paper's MMD² loss: JAX Gurvits vs MerLin photonic_engines routes (cuda_l4)
left: raw time of one loss estimate · right: JAX speedup over each route at equal precision (SE = 1% of |L|; > 1: JAX faster)
photon sweep at m = 32
seconds per single loss estimate (raw)
photons n
photon sweep at m = 32
JAX speedup over route (equal precision)
photons n
mode sweep at n = 4
seconds per single loss estimate (raw)
modes m
mode sweep at n = 4
JAX speedup over route (equal precision)
modes m
JAX Gurvits (this reproduction)
photonic_engines Gurvits estimator
photonic_engines SLOS (exact)
photonic_engines Clifford sampler
JAX Gurvits: forward + gradient
photonic_engines SLOS: forward + gradient